

Right triangle MTH has vertices at $M(3, -8)$, $T(8, 4)$, $H(8, -8)$. Right triangle DPT has vertices at $D(-3, -4)$, $P(-5, -6)$, $T(-3, -6)$.

1. Determine if $\triangle MTH$ and $\triangle DPT$ are congruent, similar, or neither.

2. What definition, theorem, or property justifies your conclusions?

3. Describe the transformations that would show that the figures are congruent or that they are similar. State which image you used as the pre-image.

Determine if the given polygons are congruent, similar, or neither.

4. Trapezoid $TRPS$: $T(1, 1)$, $R(4, 5)$, $P(7, 5)$, $S(10, 1)$
Trapezoid $RTLF$: $R(3, 3)$, $T(12, 15)$, $L(21, 15)$, $F(30, 3)$

 - ☐ Congruent
 - ☐ Similar
 - ☐ Neither

5. Kite $GHKM$: $G(0, 0)$, $H(1, -4)$, $K(3, -3)$, $M(4, -1)$.
Kite $RSTV$: $R(0, 0)$, $S(2, -8)$, $T(6, -6)$, $V(8, -2)$.

 - ☐ Congruent
 - ☐ Similar
 - ☐ Neither

6. Rhombus $DEFG$: $D(6, 4)$, $E(-1, 0)$, $F(-2, -8)$, $G(5, -4)$
Rhombus $BKMH$: $B(3, 2)$, $K(-0.5, 0)$, $M(-1, -4)$, $H(2.5, -2)$

 - ☐ Congruent
 - ☐ Similar
 - ☐ Neither

Lane and Tonymy are determining if parallelograms $HMTY$ and $NRPX$ are congruent, similar, or neither. Parallelogram $HMTY$ has vertices at $H(5,6)$, $M(-3,3)$, $T(-4,-1)$, and $Y(4,2)$. Parallelogram $NRPX$ has vertices at $N(10,12)$, $R(-6,6)$, $P(-8,-2)$, and $X(8,4)$. Their work is shown.

Lane	Tonymy
Parallelograms $HMTY$ and $NRPX$ are similar because $\frac{MH}{RN} = \frac{XR}{PN}$	Parallelograms $HMTY$ and $NRPX$ are not similar because $\frac{MH}{RN} = \frac{XN}{PR}$

- Who do you agree with?
- Explain the mistake that was made and how to correct it.